Ivan Fang

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Education

University of California, Los Angeles — Los Angeles, California
Bachelor of Science in Computer Science, GPA: 3.78 — *Expected Graduation: June 2028*

- **Relevant Coursework:** Programming in C++, Software Construction, Intro to Algorithms and Complexity
- **Honors:** Dean's List Fall 2025, Samueli Foundation Engineering Undergraduate Scholarship

Technical Skills

Languages: C++, Python, SQL (Postgres), JavaScript, Java, TypeScript, HTML/CSS
Frameworks: React, Node.js, Next.js, Express.js
Databases & Tools: AWS (Lambda, SQS, SNS, CDK, OpenSearch), MongoDB, Git, Docker, Playwright, Prisma

Experience

Amazon (AWS Marketplace Search) — Seattle, WA
Software Development Engineer Intern — *June 2026 – September 2026*

- Implemented an event-driven search-indexing pipeline in Java, replacing a scheduled batch rebuild-and-flip workflow for a 350K-document catalog, retiring two legacy indexing services, cutting per-region build-out from weeks of provisioning to a single deploy, and reducing update staleness from 60 minutes to near real-time
- Built a concurrency-safe OpenSearch writer using optimistic concurrency control via external versioning, ensuring concurrent real-time and bulk-backfill writers cannot overwrite fresher data or resurrect deleted records
- Provisioned a multi-AZ OpenSearch domain and its 88-field index with two 1,024-dimension kNN vector fields for semantic search using AWS CDK with a CloudFormation custom resource templated across 5 regions
- Developed a SQS-driven AWS Lambda consumer with dead-letter queues, partial-batch failure recovery, and idempotent processing; refactored a reusable catalog-fetching library shared across multiple indexing services

Intelligence Cubed — United States
Software Engineering Intern — *March 2026 – June 2026*

- Led a full frontend refactor from vanilla JavaScript to React + Tailwind CSS (shadcn/ui), establishing a unified design system and improving UI consistency across 10+ application modules
- Designed and implemented a centralized design token system (colors, spacing, typography), eliminating 11 legacy CSS files and enforcing scalable styling conventions
- Migrated complex UI flows (chat system, modals, navigation) to component-based architecture, improving maintainability and reducing UI-related technical debt

Archimedes Holdings — Singapore, Singapore
Software Engineering Intern — *May 2025 – August 2025*

- Integrated Playwright end-to-end tests into CI pipelines to automate checks, ensuring application stability
- Engineered a data synchronization service to integrate Shopify's GraphQL API with Shopee's REST API, automating inventory management across 200+ SKUs and eliminating the need for manual reconciliation

Projects

UCLA MentorSEAS — Los Angeles, California
Tech Lead — *March 2026 – Present*

- Lead a full website revamp for MentorSEAS, UCLA's engineering mentorship program serving 1100+ students annually
- Built a Python matching tool assigning 1,144 incoming students to 119 mentors across 11 majors, prioritizing same-major pairings and balancing group sizes to 7.7–11.0 students per mentor

LionCity Tutors — Singapore, Singapore
Founder — *June 2025 – Present*

- Launched a full-stack tuition marketplace serving 400+ tutors, processing 100+ matching requests in 3 months by building a scalable REST API with Express.js and MongoDB
- Built an automated Node.js newsletter pipeline delivering real-time job alerts to 400+ tutors via Sender.net
- Developed a Telegram bot to automate profile synchronization and updates, eliminating manual data entry and streamlining communication for the active tutor base